Model: HasFactory, primary key and fillable

Model: HasFactory, primary key and fillable

app\Models\Product.php

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class Product extends Model
{
    use HasFactory;

    protected $primaryKey = 'product_code';

    protected $fillable = [
        'name',
        'category',
        'price',
        'quantity',
        'supplier',
        'status',
        'description',
    ];
}
```

Seeder: five samples and ten factory products

Seeder: five samples and ten factory products

database\seeders\ProductSeeder.php

```php
<?php

namespace Database\Seeders;

use App\Models\Product;
use Illuminate\Database\Seeder;

class ProductSeeder extends Seeder
{
    public function run(): void
    {
        $products = [
            ['Acer Aspire 5 Laptop', 'Laptop', 34995.00, 12, 'Acer Philippines', 'Available', '15.6" FHD, Core i5, 8GB RAM, 512GB SSD'],
            ['ASUS VivoBook 15', 'Laptop', 29995.00, 4, 'ASUS Philippines', 'Low Stock', '15.6" FHD, Ryzen 5, 8GB RAM, 512GB SSD'],
            ['Kingston Fury 16GB DDR4', 'RAM', 2450.00, 30, 'Kingston Technology', 'Available', '3200MHz desktop memory'],
            ['Crucial MX500 500GB SSD', 'Storage', 3200.00, 25, 'Crucial', 'Available', '2.5" SATA solid state drive'],
            ['Samsung 980 1TB NVMe', 'Storage', 5400.00, 3, 'Samsung Electronics', 'Low Stock', 'M.2 NVMe PCIe 3.0 SSD'],
        ];

        foreach ($products as $p) {
            Product::create([
                'name' => $p[0],
                'category' => $p[1],
                'price' => $p[2],
                'quantity' => $p[3],
                'supplier' => $p[4],
                'status' => $p[5],
                'description' => $p[6],
            ]);
        }

        Product::factory()->count(10)->create();
    }
}
```

DatabaseSeeder: ProductSeeder registered

DatabaseSeeder: ProductSeeder registered

database\seeders\DatabaseSeeder.php

```php
<?php

namespace Database\Seeders;

use Illuminate\Database\Seeder;

class DatabaseSeeder extends Seeder
{
    public function run(): void
    {
        $this->call(ProductSeeder::class);
    }
}
```

Factory: definition()

Factory: definition()

database\factories\ProductFactory.php

```php
<?php

namespace Database\Factories;

use App\Models\Product;
use Illuminate\Database\Eloquent\Factories\Factory;

/**
 * @extends Factory<Product>
 */
class ProductFactory extends Factory
{
    /**
     * Define the model's default state.
     *
     * @return array<string, mixed>
     */
    public function definition(): array
    {
        $product = fake()->randomElement([
            ['Wireless Mouse', 'Peripherals', 500, 3000],
            ['Mechanical Keyboard', 'Peripherals', 1500, 7000],
            ['Gaming Monitor', 'Monitor', 8000, 25000],
            ['USB-C Hub', 'Peripherals', 800, 5000],
            ['Portable SSD', 'Storage', 2500, 12000],
            ['Wi-Fi Router', 'Networking', 1500, 10000],
        ]);
        $quantity = fake()->numberBetween(0, 30);

        return [
            'name' => $product[0].' '.fake()->bothify('??-###'),
            'category' => $product[1],
            'price' => fake()->randomFloat(2, $product[2], $product[3]),
            'quantity' => $quantity,
            'supplier' => fake()->company(),
            'status' => $quantity === 0 ? 'Out of Stock' : ($quantity <= 5 ? 'Low Stock' : 'Available'),
            'description' => fake()->sentence(),
        ];
    }
}
```

Terminal: successful migrate:fresh --seed

Terminal: successful migrate:fresh --seed

```
PS C:\Users\Desca\techzone> php artisan migrate:fresh --seed --no-interaction

 Dropping all tables .. 58.54ms DONE

 INFO Preparing database.

 Creating migration table .. 12.53ms DONE

 INFO Running migrations.

 0001_01_01_000000_create_users_table .. 50.97ms DONE
 0001_01_01_000001_create_cache_table .. 45.64ms DONE
 0001_01_01_000002_create_jobs_table .. 50.57ms DONE
 2026_10_07_152645_create_products_table .. 5.15ms DONE


 INFO Seeding database.

 Database\Seeders\ProductSeeder .. RUNNING
 Database\Seeders\ProductSeeder .. 46 ms DONE
```

Database: product_code primary key, 15 rows

Database: product_code primary key, 15 rows

```
mysql> SHOW COLUMNS FROM products LIKE product_code; SELECT product_code, name, category, quantity, status FROM products ORDER BY product_code;

+--------------+---------------------+------+-----+---------+----------------+
| Field        | Type                | Null | Key | Default | Extra          |
+--------------+---------------------+------+-----+---------+----------------+
| product_code | bigint(20) unsigned | NO   | PRI | NULL    | auto_increment |
+--------------+---------------------+------+-----+---------+----------------+
+--------------+-------------------------+-------------+----------+-----------+
| product_code | name                    | category    | quantity | status    |
+--------------+-------------------------+-------------+----------+-----------+
|            1 | Acer Aspire 5 Laptop    | Laptop      |       12 | Available |
|            2 | ASUS VivoBook 15        | Laptop      |        4 | Low Stock |
|            3 | Kingston Fury 16GB DDR4 | RAM         |       30 | Available |
|            4 | Crucial MX500 500GB SSD | Storage     |       25 | Available |
|            5 | Samsung 980 1TB NVMe    | Storage     |        3 | Low Stock |
|            6 | Gaming Monitor ye-548   | Monitor     |        9 | Available |
|            7 | Portable SSD ji-009     | Storage     |        4 | Low Stock |
|            8 | Portable SSD pe-474     | Storage     |        5 | Low Stock |
|            9 | Gaming Monitor xv-300   | Monitor     |       23 | Available |
|           10 | USB-C Hub ll-750        | Peripherals |       15 | Available |
|           11 | Gaming Monitor gc-178   | Monitor     |       30 | Available |
|           12 | Gaming Monitor sw-335   | Monitor     |        9 | Available |
|           13 | Wi-Fi Router xo-653     | Networking  |       10 | Available |
|           14 | Wi-Fi Router xt-590     | Networking  |        2 | Low Stock |
|           15 | Wireless Mouse ol-861   | Peripherals |       25 | Available |
+--------------+-------------------------+-------------+----------+-----------+
+----------------+
| total_products |
+----------------+
|             15 |
+----------------+
```

Browser: 15 generated product codes

TECHZONE INVENTORY

← Back

Products

15 items in inventory

+ Add Product

Search by code, name, or supplier...

All categories

All statuses

CODE	NAME	CATEGORY	PRICE	QTY	SUPPLIER	STATUS	ACTIONS
1	Acer Aspire 5 Laptop	Laptop	₱34,995.00	12	Acer Philippines	Available	
2	ASUS VivoBook 15	Laptop	₱29,995.00	4	ASUS Philippines	Low Stock	
3	Kingston Fury 16GB DDR4	RAM	₱2,450.00	30	Kingston Technology	Available	
4	Crucial MX500 500GB SSD	Storage	₱3,200.00	25	Crucial	Available	
5	Samsung 980 1TB NVMe	Storage	₱5,400.00	3	Samsung Electronics	Low Stock	
6	Gaming Monitor ye-548	Monitor	₱20,660.58	9	Crooks, Lemke and Herzog	Available	
7	Portable SSD ji-009	Storage	₱4,609.75	4	Zboncak Inc	Low Stock	
8	Portable SSD pe-474	Storage	₱8,563.94	5	Murray-Mills	Low Stock	
9	Gaming Monitor xv-300	Monitor	₱12,194.32	23	Kulas, Daniel and Romaguera	Available	
10	USB-C Hub ll-750	Peripherals	₱1,898.65	15	Kub-Bayer	Available	
11	Gaming Monitor gc-178	Monitor	₱11,093.20	30	Cruickshank-Hahn	Available	
12	Gaming Monitor sw-335	Monitor	₱12,529.47	9	O'Connell, Swift and Spencer	Available	
13	Wi-Fi Router xo-653	Networking	₱6,470.35	10	Zulauf, Nienow and Haag	Available	
14	Wi-Fi Router xt-590	Networking	₱4,564.81	2	Corkery, Fay and Upton	Low Stock	
15	Wireless Mouse ol-861	Peripherals	₱2,688.88	25	McClure, Bednar and Abshire	Available	